Md Monirul Islam

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Education

- **Bangladesh University of Engineering and Technology (BUET)** Dhaka, Bangladesh
  B.Sc. in Electrical and Electronic Engineering *March 2018 – May 2023*
  CGPA: 3.85/4.00 **(3.98/4.00 in the final year)**, ranked **25 of 189**, awarded with Honours

  B.Sc. Thesis: α-Graphyne ballistic nanoribbon FET — quantum-transport modeling (advisor: Dr. Mahbub Alam)

Deployed Systems

• **3-robot AGV fleet** — live production at The Urmi Group RMG facility, Dhaka (2025 – present). • **EOD tracked ROVs (Jontro Soinik v1 / v2)** — fielded in UN peacekeeping / humanitarian-demining support (Mali 2024, Congo 2025).

Research Interests

• Multi-Robot Coordination & Motion Planning • Path Planning under Uncertainty • State Estimation & Sensor Fusion • Real-Time Embedded Control • Field Deployment of Autonomous Systems in Human-Shared Spaces

Research Experience

- **Distributed multi-AGV path coordination — deployed in production.** Coordination scheme for a 3-robot fleet with **months of zero-collision operation** in a live human-occupied factory: each robot plans its own A* path and publishes timestamped grid waypoints that peers treat as spatiotemporal reservations (vertex / swap / target conflict checks, continuous timestamp maintenance), while a fail-closed coordinator (*FleetCore*) handles job dispatch, mutual exclusion, and deadlock recovery.
- **GA-tuned bilateral cross-coupled BLDC speed synchronization.** Two mismatched BLDC motors, each on its own MCU; a bilateral cross-coupled term evaluated identically on both nodes from a dedicated inter-MCU speed exchange drives the inter-motor speed difference toward zero, with gains tuned offline against an identified motor model. Outperforms an independent-PID baseline on transient and steady-state synchronization.
- **Latency-compensated 5-state EKF for visual-fiducial localization.** Fuses wheel-encoder odometry with AprilTag fixes; velocity-bias states absorb systematic encoder drift over long corridors, and a rewind-and-replay step applies each camera fix at its true measurement timestamp. Drift-free hand-off between moving (Kalman) and parked (complementary) modes.
- **Closed-form analytical IK for a tracked mobile manipulator.** From-scratch analytical solver replacing MoveIt's KDL with deterministic, branch-stable solutions; smooth turret-scaling preserves continuity through the home-pose singularity and an FK-anchored accumulator suppresses un-commanded cross-axis motion. Platform combines a 6-DOF arm, a 2-DOF anti-flipper arm (also a collision-avoidance element), and a differential-drive base.
- **Fault-tolerant dual-network feedback for multi-node robotic platforms.** Two redundant feedback networks carry coordination data in parallel; the receiver detects channel degradation and fails over to the healthy network without losing a control cycle — targeting single-point-of-failure risks that surface in field deployment.
- **Battery-sizing optimizer for a high-payload multirotor.** Paper-stage design and aerodynamic simulation of a heavy-lift multirotor, plus an in-house optimizer that computes feasible battery-pack configurations (hover-throttle reachability, per-motor current limits, voltage compatibility, cruise margin) from payload, flight-time, and motor parameters. Published as an interactive client-side tool.

Publications

Peer-reviewed

- Md Shakil Tanvir, Fiaz Ahmed Tonmoy, **Md Monirul Islam**, Mainul Islam, Mahamudul Hasan, Raihan Ur Rashid Razeen, "Optimizing Master-Slave Broadcast Communication in Multi-Node Networks Using RS485 Standard", 27^{th} International Conference on Computer and Information Technology (ICCIT), IEEE, 2024. doi: 10.1109/ICCIT64611.2024.11022430
- H. Khan, **Md Monirul Islam**, R.I. Roya, S.N. Azad, M. Alam, "Performance Analysis of an α-Graphyne Nano-Field Effect Transistor", *Micromachines*, MDPI, 2023. doi: 10.3390/mi14071385 *(first four authors contributed equally)*

Manuscripts submitted / in preparation

- **Md Monirul Islam**, Mainul Islam, Md Shakil Tanvir, "Speed Synchronization of a Dual-BLDC Tracked Differential-Drive Vehicle Using a GA-Tuned Multi-Rate Cross-Coupled Controller", Submitted, 2026 (first / corresponding author).
- Md Shakil Tanvir, **Md Monirul Islam**, Fiaz Ahmed Tonmoy, Mainul Islam, "A Dual-Network Feedback Communication System for Multi-Node Unmanned Robotic Control", Submitted, 2026 (co-author).
- **Md Monirul Islam**, Mainul Islam, Emon Hasan Saumik, "Peer-Aware Distributed Path Planning for a Production-Deployed Multi-AGV Fleet in a Human-Occupied Garment Factory", In Preparation, 2026 (first / corresponding author).

Professional Experience

- **Research and Development Unit, Cybernetics Hi-Tech Solutions (Pvt) Ltd** Dhaka, Bangladesh
  R&D Engineer and Team Lead *(June 2023 – present)*
  - **CyberFleet (production AGV fleet).** Lead engineer end-to-end. The platform productised the work and is now operating continuously at The Urmi Group RMG facility, Dhaka.
  - **Jontro Soinik v1 / v2 EOD ROVs.** Owned the electrical / firmware side across both generations — schematic design, in-house PCB, power distribution, and on-vehicle firmware; ran operator training and field hand-over to receiving units. (Deployment context under *Deployed Systems.*)
  - **Jontro Soinik 2.0 (6-DOF mobile manipulator).** Owned the control / software stack for the tracked tele-op platform.
  - **SD15-N target-tracking surveillance drone.** Designed the autonomous-tracking pipeline (vision → MAVLink → Pixhawk).
  - **Titan 550 industrial CoreXY 3D printer.** Two-iteration product build; the in-house fab loop that produces fixtures, brackets, and prototypes for every other platform on this list.
  - **In-house PCB & CNC.** Designed all controller / communication boards across the ROV & AGV platforms in Altium / KiCad and milled them in-house on a 3-axis CNC — no external fab. Also: hybrid VTOL quad-plane prototype; real-time pet-food sorting line for an industrial-automation client.
- **Ulkasemi Pvt. Limited** Dhaka, Bangladesh
  Intern (Dec 2022 – Mar 2023)
  - Worked on IC design using TSMC 180 nm and GPDK 90 nm technologies.
  - Developed a Wishbone bus interface in Verilog; validated a high-speed BiFET op-amp in Cadence Virtuoso.

Selected Undergraduate Projects

- **5-DOF robotic arm (analytical IK)**: Designed in SolidWorks, 3D-printed on a self-built hobby printer using custom 3D-printed planetary gearboxes; closed-form analytical inverse-kinematics control on stepper motors.
- **Autonomous GPS-waypoint delivery drone**: Mechanical design in SolidWorks, Arduino flight controller with PID stabilization, GPS-waypoint navigation, and a custom payload-deployment mechanism.
- **RV32I multicycle RISC-V processor + Wishbone memory controller**: Custom Verilog implementation with full testbench verification; physical-layout exploration in Cadence at TSMC 180 nm.
- **Regulated boost converter (TSMC 180 nm, Cadence)**: Schematic capture and physical layout of a voltage-regulated DC-DC boost converter.

Skills Summary

- **Robotics & Middleware**: ROS 2 (Jazzy), MoveIt 2, ros2_control, JointTrajectoryController, URDF / Xacro / SRDF, AprilTag, MAVLink, dronekit, pymavlink
- **Embedded & MCU**: STM32 (HAL & LL), ESP32, AVR, Arduino, ARM Cortex-M, Keil µVision, BigTreeTech 32-bit boards
- **Control & Algorithms**: Closed-form analytical IK, Extended Kalman Filter, PID (cascaded, GA-tuned), S-curve motion profiling, distributed A* path planning, peer-aware spatiotemporal reservations
- **Flight, Vision & Simulation**: ArduPilot, iNav, Betaflight, Pixhawk, YOLOv11n, OpenCV; MATLAB / Simulink, Cadence Virtuoso, PSPICE, Icarus Verilog, GTKWave
- **PCB, CAD & Fabrication**: Altium Designer, KiCad, Autodesk Eagle, Proteus; SolidWorks; in-house 3-axis CNC milling; Marlin, Ultimaker Cura, FlatCAM, GRBL
- **Programming & Docs**: Python, C, Embedded C, Verilog; LaTeX, Jekyll (al-folio), Microsoft Office

Honors and Awards

- University Merit Scholarship, BUET — three times (Jan 2019, Jan 2020, Jan 2022), for outstanding academic performance
- Dean's List Award, BUET — every academic year (GPA 3.75+)
- University Stipend, BUET — three times (Jan 2018, Jul 2018, Jul 2021)

References

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